

Problem Set 1
Winter 2020

Issued: Thursday, January 9

Due: Thursday, January 16

Notation: The symbols $\{\xrightarrow{d}, \xrightarrow{p}, \xrightarrow{m.s.}\}$ denote convergence in distribution, probability and mean-square respectively.

Probability background: Keener Chapter 1 and Sections 6.1, 6.2, 8.1, and 8.2.

Reading: Sections 3.1–3.3

Full credit: 100 points.

Problem 1.1 (20 points)

Suppose X_i is a sequence of independent random variables with a common mean $\mathbb{E}(X_i) = \mu$ for all i and different variances $\mathbb{E}(X_i - \mu)^2 = \sigma_i^2$.

- (a) Show that $\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n} \xrightarrow{m.s.} \mu$ as long as $\sum_{i=1}^n \sigma_i^2 = o(n^2)$. Under the same assumptions, does $\bar{X}_n \xrightarrow{p} \mu$? Why or why not?
- (b) Now consider the estimate $\tilde{X}_n = \frac{\sum_{i=1}^n \frac{1}{\sigma_i} X_i}{\sum_{i=1}^n \frac{1}{\sigma_i}}$. Prove that $\text{var}(\tilde{X}_n) \leq \text{var}(\bar{X}_n)$. Use this to conclude that, under the same conditions as in part (a), $\tilde{X}_n \xrightarrow{p} \mu$.

Hint: Use Cauchy–Schwarz inequality to deduce: $n^2 \leq (\sum \frac{1}{\sigma_i})(\sum \sigma_i)$ and $\sum \sigma_i \leq \sqrt{n}(\sum \sigma_i^2)^{1/2}$.

Problem 1.2 (15 points)

Consider an i.i.d. sample $\{X_1, \dots, X_n\}$ from the uniform distribution on $[0, \theta]$, and the estimator $M_n = \max\{X_1, \dots, X_n\}$.

- (a) Prove that $M_n \xrightarrow{p} \theta$ as $n \rightarrow +\infty$.
- (b) Compute the bias and variance of the estimator M_n as a function of θ .
- (c) Compute the risk of M_n under quadratic loss.

Problem 1.3 (40 points)

Consider the model $X \sim N(\theta, 1)$ for $\theta \in \mathbb{R}$.

- (a) Compute the PDF of X^2 ; call it $p_{X^2}(t)$.
- (b) Obtain the conditional distribution of X given $X^2 = t$ for $t \geq 0$.

Hint: This is a discrete distribution. See part (c) to guide your choice of the weights.

- (c) Verify your answer in part (b), by showing that

$$\int \mathbb{P}(X \in I \mid X^2 = t) p_{X^2}(t) dt = \mathbb{P}(X \in I),$$

for any interval $I = (-\infty, a] \subset \mathbb{R}$.

Hint: Consider $a > 0$ and $a < 0$ separately. For each case, differentiate the two sides of the identity w.r.t to a .

- (d) Based on part (b), is X^2 a sufficient statistic in this model?

Problem 1.4 (15 points)

A parameterization $\theta \mapsto \mathbb{P}_\theta$ is identifiable if $\theta_1 \neq \theta_2$ implies that $\mathbb{P}_{\theta_1} \neq \mathbb{P}_{\theta_2}$. Which of the following parameterizations are identifiable?

- (a) Suppose that $X_1, \dots, X_p$ are independent with $X_i \sim \mathcal{N}(\alpha_i + \nu, \sigma^2)$. Set

$$\theta = (\alpha_1, \dots, \alpha_p, \nu, \sigma^2)$$

and consider the family of joint distributions $\mathbb{P}_\theta$ over $(X_1, \dots, X_p)$.

- (b) Suppose that X is a Bernoulli variable, and we parameterize its probability mass function with $\theta = (\theta_0, \theta_1)$ and $p(x; \theta) \propto \exp(\theta_0 x + \theta_1(1 - x))$.
- (c) Autoregressive process: consider the collection of RVs $(X_1, \dots, X_p)$ given by $X_1 \sim \mathcal{N}(0, \sigma^2)$, and $X_{i+1} = \alpha X_i + \sqrt{1 - \alpha^2} W_i$ where $W_i \sim \mathcal{N}(0, \sigma^2)$, independent of the X_i , and $\alpha \in [0, 1]$. Let $\theta = (\alpha, \sigma^2)$ and let $\mathbb{P}_\theta$ index the joint distribution over $(X_1, \dots, X_p)$.

Problem 1.5 (20 points)

Keener: Problem 1 from Chapter 3, parts (a)–(d).

Problem 1.6 (20 points)

Suppose that $(X_1, \dots, X_n)$ are i.i.d. Poisson random variables with parameter θ . Show that $T = \sum_{i=1}^n X_i$ is sufficient in two ways:

- (a) first use direct methods: compute the conditional distribution given $T = t$.
- (b) apply the factorization theorem.

Problem 1.7 (15 points)

Consider a nonparametric regression model $y_i = f(x_i) + \varepsilon_i$, $i = 1, \dots, n$, where $f : [0, 1] \rightarrow \mathbb{R}$ is some unknown function, $\{x_i, i = 1, \dots, n\} \subset [0, 1]$ are known deterministic points and $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. We would like to estimate f with a *stump*, i.e., a function of the form

$$f_{\tau, \beta}(x) = \beta_1 1\{x < \tau\} + \beta_2 1\{x > \tau\}.$$

Let us define the estimator as

$$\hat{f}_\tau := f_{\tau, \hat{\beta}}, \quad \text{where} \quad \hat{\beta} := \arg \min_{\beta \in \mathbb{R}^2} \sum_{i=1}^n (y_i - f_{\tau, \beta}(x_i))^2.$$

Consider the loss function $L(f, \hat{f}_\tau) := \|f - \hat{f}_\tau\|_{L^2}^2 := \int_0^1 (f(x) - \hat{f}_\tau(x))^2 dx$, so that the risk $R(f, \hat{f}_\tau) = \mathbb{E}[L(f, \hat{f}_\tau)]$ is the integrated mean-squared error. (You can safely interchange $\mathbb{E}$ and $\int_0^1$ in this problem.)

- (a) Derive a bias-variance decomposition for the risk.
- (b) Evaluate the risk for $\hat{f}_\tau$ using the decomposition of part (a).
- (c) Assume $x_i = i/n$ for $i = 1, \dots, n$. Find asymptotic expressions for the bias-term and n times the variance-term of the risk. That is, if $R = B_n^2 + V_n$ is the bias-variance decomposition, find $\lim B_n^2$ and $\lim nV_n$ as $n \rightarrow \infty$.